Installation instructions

Install ROS2 Humble

Ubuntu (Debian packages)

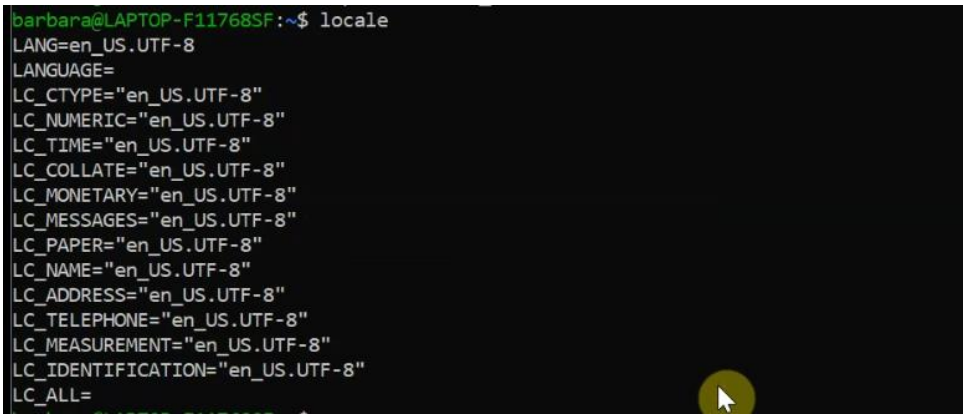
Set locale

Write in the Linux terminal:

```
locale  # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # verify settings
```



Setup Sources

Write in the Linux terminal:

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

```
sudo apt update && sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg
```

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

Install ROS 2 packages

Write in the Linux terminal:

```
sudo apt update
sudo apt upgrade
```

```
sudo apt install ros-humble-desktop
```

```
sudo apt install ros-humble-ros-base
```

```
sudo apt install ros-dev-tools
```

Environment setup

Write at the end of the .bashrc:

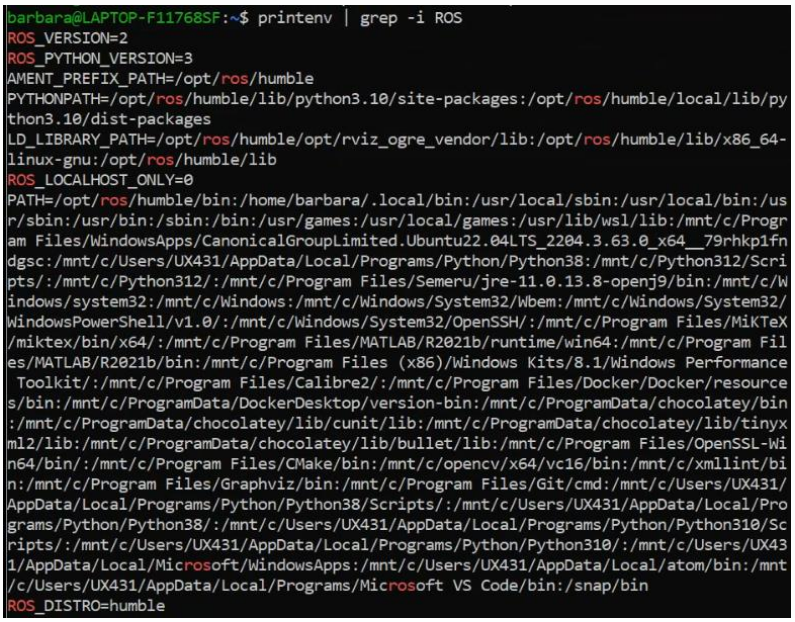
```
source /opt/ros/humble/setup.bash
```

Close and open again the Linux terminal.

Check environment variables

Write in the Linux terminal and check that: `ROS_VERSION=2` `ROS_PYTHON_VERSION=3` `ROS_DISTRO=humble`

```
printenv | grep -i ROS
```



Install SOFA

Building Sofa

Compiler

Write in the Linux terminal:

```
sudo apt install build-essential software-properties-common
```

GCC

Write in the Linux terminal:

```
apt-cache search '^gcc-[0-9.]+$'
```

Install the latest one (example with gcc-11):

```
sudo apt install gcc-11
```

Clang

Write in the Linux terminal:

```
apt-cache search '^clang-[0-9.]+$'
```

Install the latest one (example with clang-12):

```
sudo apt install clang-12
```

CMake: Makefile generator

Write in the Linux terminal:

```
sudo apt install cmake cmake-gui
```

Ninja: build system

Write in the Linux terminal:

```
sudo apt install ninja-build
```

CCache: caching system

Write in the Linux terminal:

```
sudo apt install ccache
```

Dependencies: Core (required)

tinyXML2

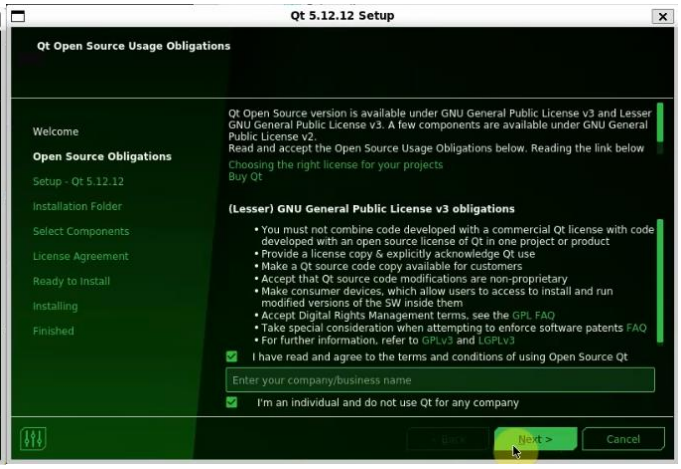
Write in the Linux terminal:

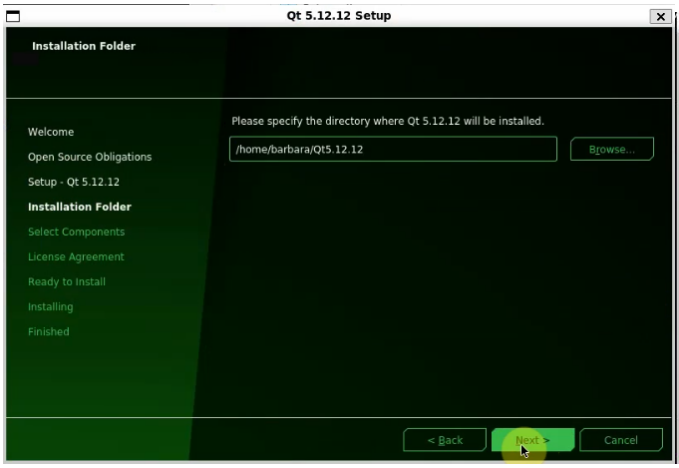
```
sudo apt install libtinyxml2-dev
```

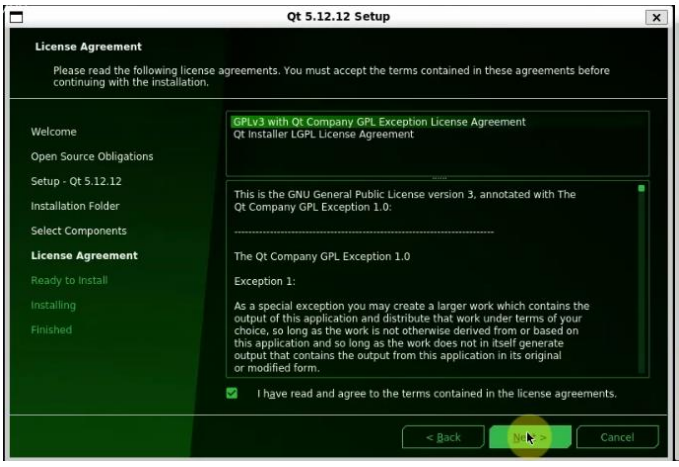
Qt

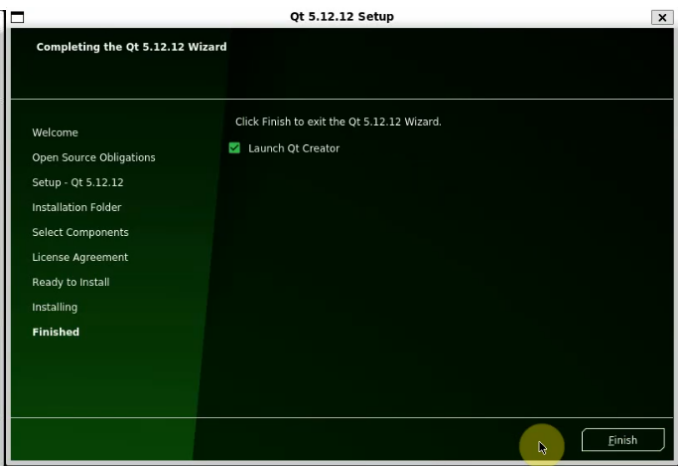
Download Qt >= 5.12.0 with **Charts** and **WebEngine.** It is recommendable to install Qt **in your user directory** with the unified installer.

https://download.qt.io/archive/qt/5.12/









OpenGL

Write in the Linux terminal:

```
sudo apt install libopengl0
```

Boost

Write in the Linux terminal:

```
sudo apt install libboost-all-dev
```

Python 3.10 + pip + numpy + scipy

Write in the Linux terminal:

```
sudo apt install python3.10-dev pybind11-dev
sudo apt-get install python3.10-distutils
sudo curl -L https://bootstrap.pypa.io/pip/get-pip.py –output /tmp/get-pip3.py
sudo python3.10 /tmp/get-pip3.py
sudo python3.10 -m pip install –upgrade pip
sudo python3.10 -m pip install numpy scipy pybind11==2.9.1
```

Additional libraries: libPNG, libJPEG, libTIFF, Glew, Zlib

Write in the Linux terminal:

```
sudo apt install libpng-dev libjpeg-dev libtiff-dev libglew-dev zlib1g-dev
```

Eigen

Write in the Linux terminal:

```
sudo apt install libeigen3-dev
```

Dependencies: Plugins (optional)

CGALPlugin

Write in the Linux terminal:

```
sudo apt install libcgal-dev libcgal-qt5-dev
```

MeshSTEPLoader

Write in the Linux terminal:

```
sudo apt install liboce-ocaf-dev
```

SofaAssimp

Write in the Linux terminal:

```
sudo apt install libassimp-dev
```

SofaCUDA

Write in the Linux terminal:

```
sudo apt install nvidia-cuda-toolkit
```

SofaHeadlessRecorder

Write in the Linux terminal:

```
sudo apt install libavcodec-dev libavformat-dev libavutil-dev libswscale-dev
```

SofaPardisoSolver

Write in the Linux terminal:

```
sudo apt install libblas-dev liblapack-dev
```

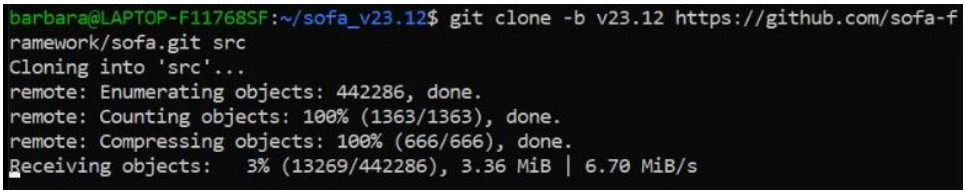
Setup your source and build directories

Create SOFA directories as follows:

```
sofa_v23.12/
├── build/
└── src/
    └── < SOFA sources here >
```

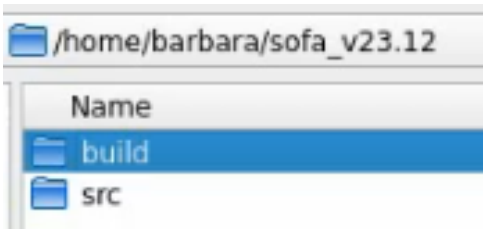
In the Linux terminal, go into sofa_v23.12/src/ folder and clone the last version (example for version v23.12):

```
git clone -b v23.12 https://github.com/sofa-framework/sofa.git
```



Generate a Makefile with CMake

Create build directories respecting the arrangement



In Linux terminal and Run CMake-GUI

```
cmake-gui
```

```
barbara@LAPTOP-F11768SF:~/sofa_v23.12/build$ cmake-gui
```

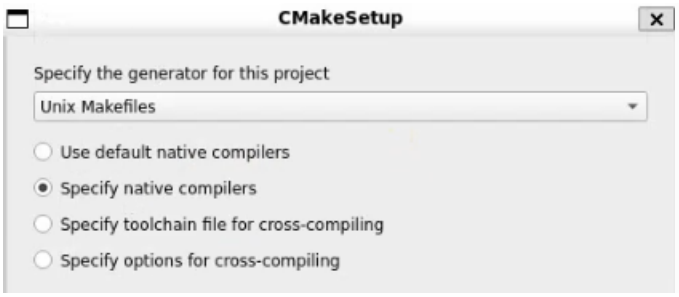
Set source folder and build folder.

Run Configure.

Select “Unix Makefile”, “Specify native compilers” and press “Next”.

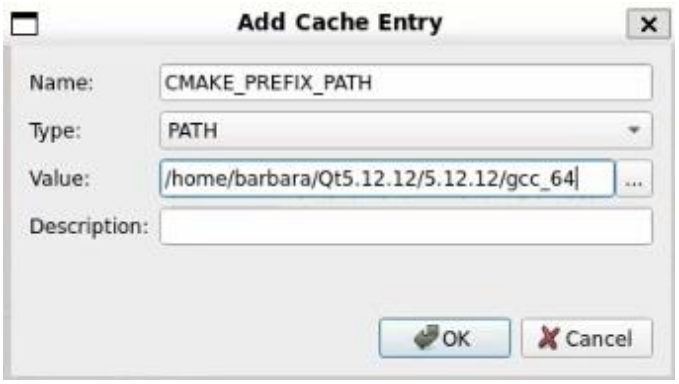
Set the C compiler to “/usr/bin/gcc” and C++ compiler to “/usr/bin/g++”.

Run Configure



Qt detection error

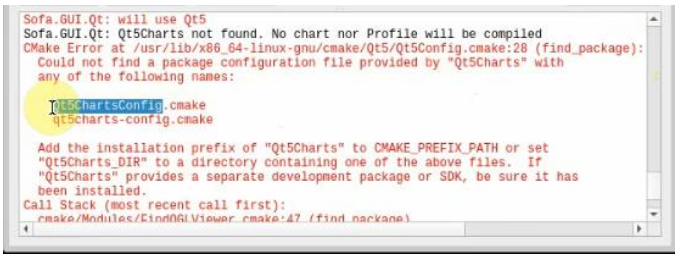
Add an entry called CMAKE_PREFIX_PATH with path /home/YOUR_USERNAME/Qt/QT_VERSION/COMPILER matching your Qt installation. Then, Configure again.

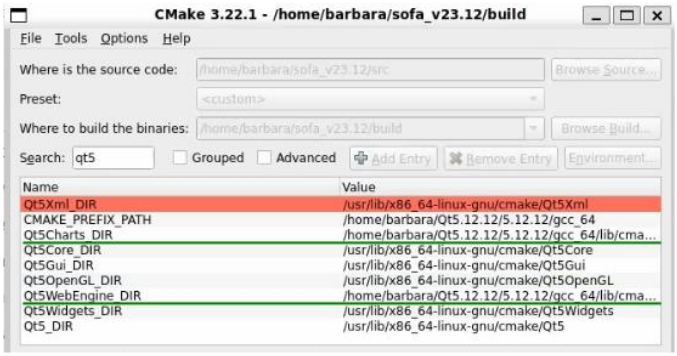


QtCharts and QtWebEngine error

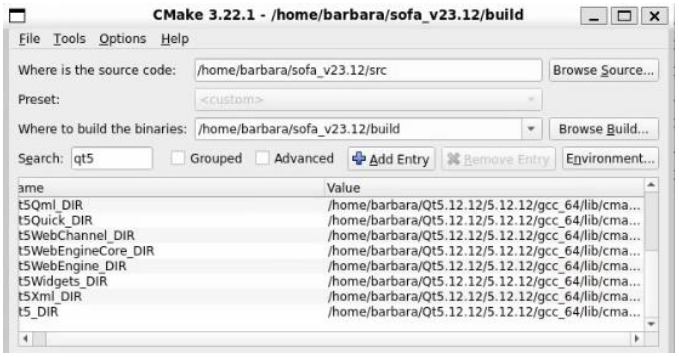
On Qt5_Charts_DIR insert the path “/home/barbara/Qt5.12.12/5.12.12/gcc_64/lib/cmake/QtCharts”. Then, Configure again.

Do the same procedure with QtWebEngine.





Do the same with all Qt5XXX_DIR that is not directed to the correct file.

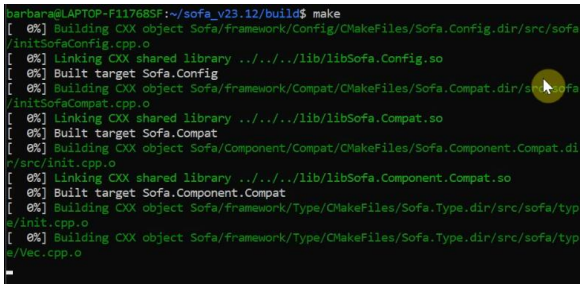


When you are ready, Configure again and run Generate

Compile

Open a terminal in your build directory and run

```
make
```



Then run Sofa

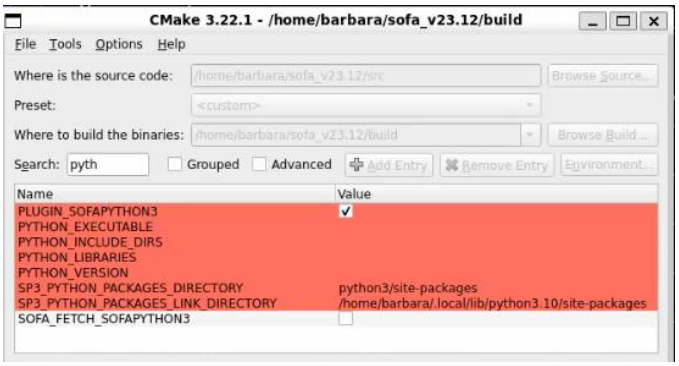
```
./bin/runSofa
```

Pugins

Python (internal)

In CMake-GUI, check the box “SOFA_FETCH_SOFAPYTHON3”. Then, Configure.

The option “PLUGIN_SOFAPYTHON3” will appear together with other things. The “PLUGIN_SOFAPYTHON3” box should be checked and the “SOFA_FETCH_SOFAPYTHON3” should be unchecked. Then, Configure again.



Finally, run Generate and on the terminal run “make” again.

```
make
```

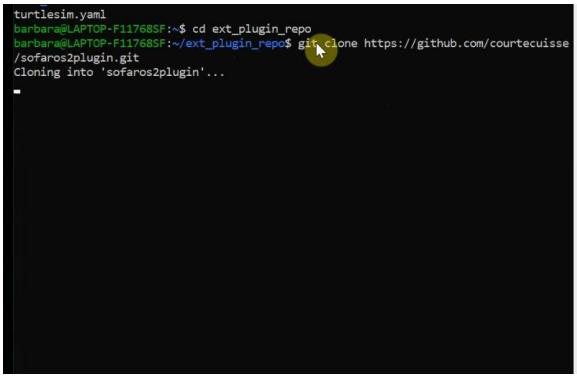
ROS2 (external)

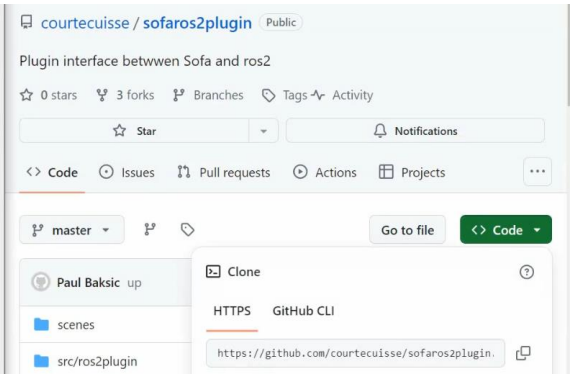
Create a folder called “ext_plugin_repo” outside of the “sofa” folder. In this directory, the structure is:

- ext_plugin_repo/
  - plugin1/
  - plugin2/
  - …

Clone the GitHub sofaros2plugin repository on the “ext_plugin_repo” folder.

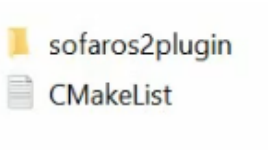
https://github.com/InfinyTech3D/sofaros2plugin



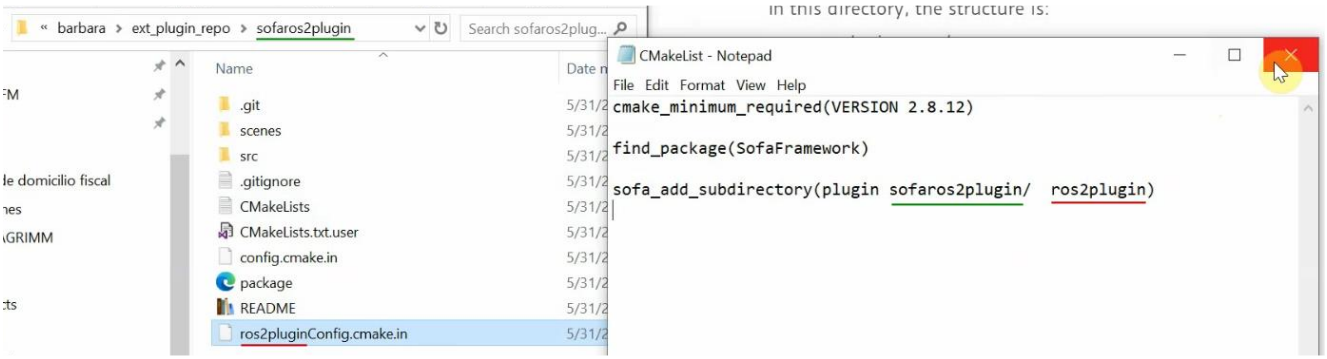


Create a CMakeList.txt and write on it as follows:

```
cmake_minimum_required(VERSION 2.8.12)
find_package(SofaFramework)
sofa_add_subdirectory(plugin path_to_plugin1/  name_of_project_plugin1)
sofa_add_subdirectory(plugin path_to_plugin2/  name_of_project_plugin2)
```



In the case of the ROS2 plugin, it will be:



To compile all the external plugins located in this repository, all you need to do is to set the path to this repository (*/ext_plugin_repo/*) in the CMake variable: **SOFA_EXTERNAL_DIRECTORIES**.

Then, Configure again. Finally, run Generate and on the terminal run “make” again.

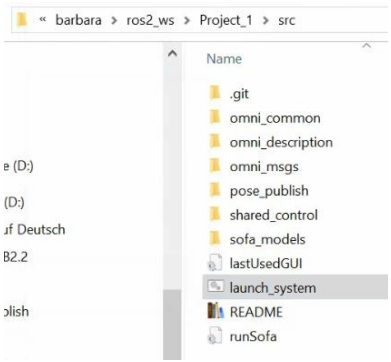
```
make
```

.bashrc

Add to the .bashrc the following lines:

Set up the project

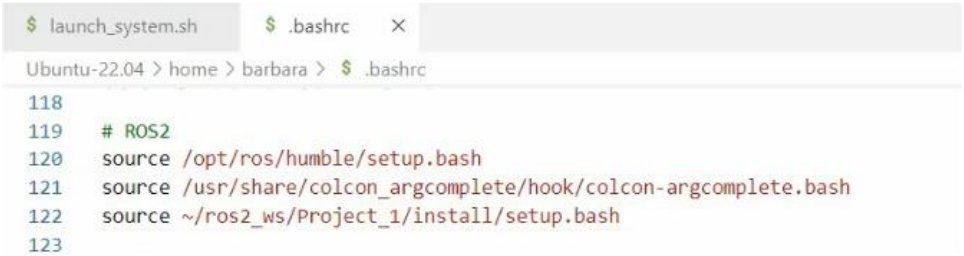
Setup your source and build directories



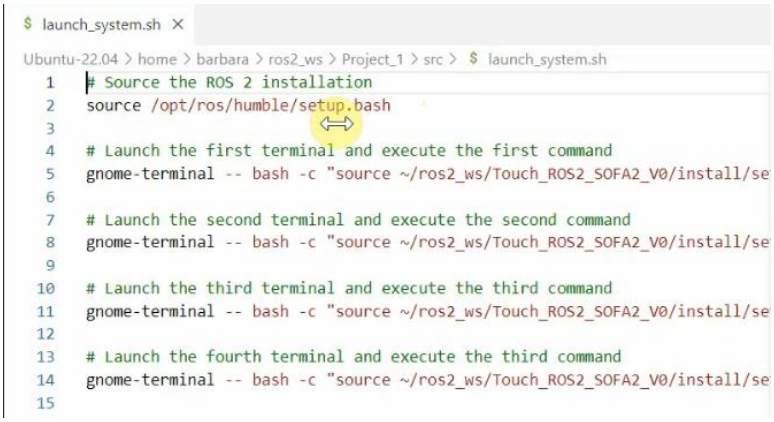
Create directories as follows inside the ros2_ws and clone the project on the src folder.

```
ros2_ws/
└── Project_1/
    └── src/
        └── < clone project here >
```

Add to the .bashrc the following lines:

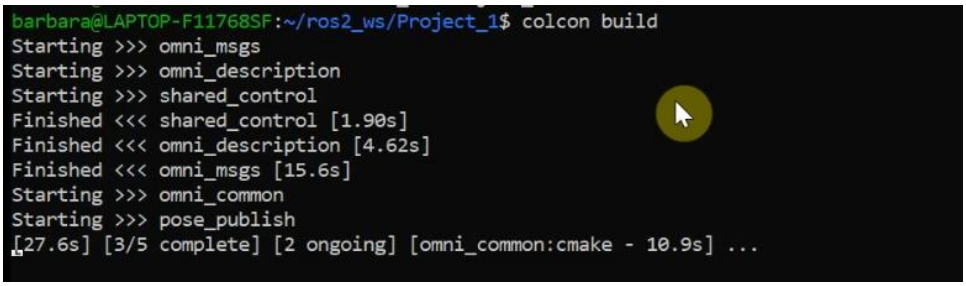


Change the .launch_system to customized it to your personal folder names:



On the terminal, on the project folder, run:

```
colcon build
```

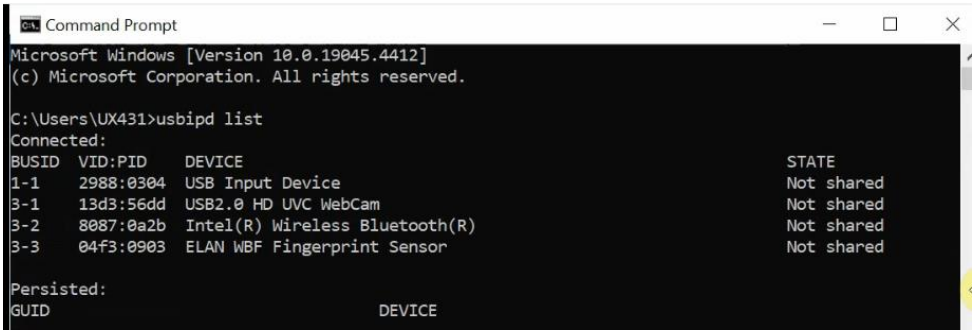


Installation of drivers

Only for WSL: Patch USB device from Win to WSL

On Windows command window run

```
usbipd list
```

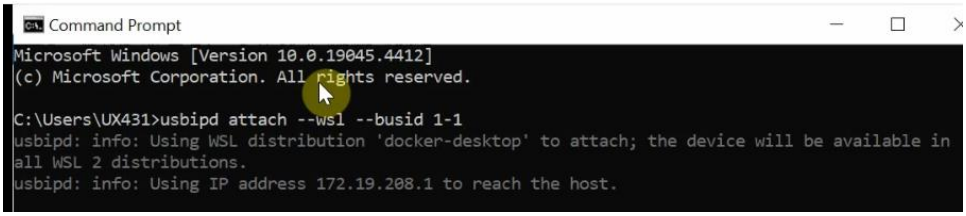


Note which BUSID corresponds to "USB Input Device" (in the example 1-1). Then, as admin run the following command:

```
usbipd bind --busid 1-1
```

Next step has to be made every time that you reconnect the haptic device. It should be run in a no admin command window:

```
usbipd attach --wsl --busid 1-1
```



Drivers installation in Linux

Add to the .bashrc:

```
export GTDD_HOME=/usr/share/3DSystems
```

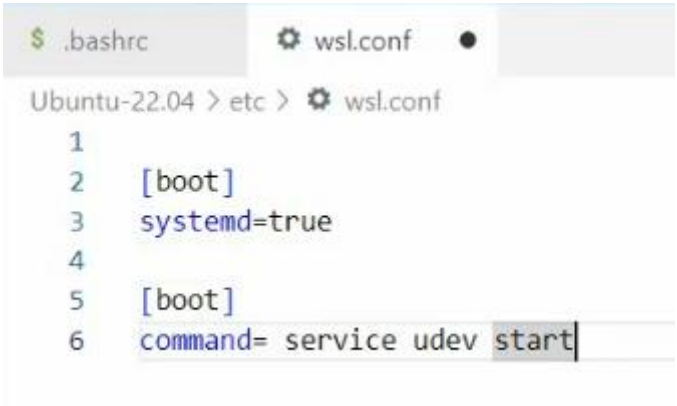
On the terminal:

```
sudo mkdir /usr/share/3DSystems/config -p
```

```
sudo chmod -R 777 /usr/share/3DSystems/
```

add to /etc/wsl.conf:

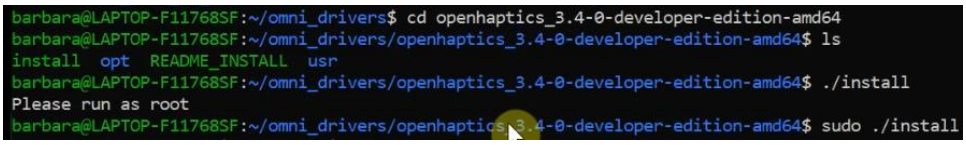
```
[boot] command= service udev start
```



On the terminal:

```
sudo usermod -a -G users [yourUSER]
```

Install openhaptics_3.4-0-developer-edition-amd64. To make this, go into the openhaptics_3.4-0-developer-edition-amd64 folder and execute the installation file as follows:



install TouchDriver_2023_01_12. To make this, go into the TouchDriver_2023_01_12 folder and execute the installation file as follows: